

Figure 6: Cosine similarity of target-token hidden states across layers in LLaMA-3-8B.

representation space. In contrast, because subject information is weakly propagated for UH samples, their last-token representations receive smaller updates and therefore are expected to remain *tightly clustered*, dominated by the shared features of the prompt template.

To verify this, we compute the layer-wise cosine similarity among last-token representations $\mathbf{h}_T^\ell$ within each knowledge category (Figure 6). The results indicate that the strength of propagated subject information significantly affects the geometric properties. In early layers, cosine similarity is high for all categories (≈ 0.9). From the mid-layers onward, where subject representations propagate for FAs and AHs, their last-token representations diverge significantly. By layer 25, the cosine similarity for both FAs and AHs drops to ≈ 0.2 . In contrast, UHs remain moderately clustered, with cosine similarity only declining to ≈ 0.5 .

To further illustrate this, we present a t-SNE visualization of the last token’s representations at layer 25 of LLaMA-3-8B in Figure 7. The visualization aligns with our cosine similarity analysis: hidden representations of UHs form a distinct cluster separated from FAs, whereas AHs substantially overlap with FAs. We quantitatively validate this observation using two cluster separability metrics (the Silhouette coefficient and Davies-Bouldin Index) in Appendix D.2, which show significantly higher separability for UHs compared to AHs. Additional visualizations can be found in Appendix D.3.

Additionally, we observe that the strength of subject representation propagation also affects the entropy of the output distribution (Figure 8). A strong subject-specific update concentrates the output distribution on specific subject-related entities, resulting in low-entropy outputs for FAs and AHs. Conversely, a weak subject-specific update leads to a wider spread of probability mass covering many possible answers (e.g., assigning probability to multiple possible names for the prompt “The name of

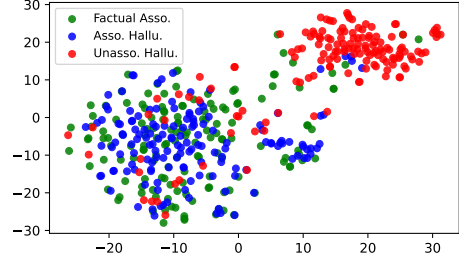


Figure 7: t-SNE visualization of last token’s representations at layer 25 of LLaMA-3-8B.

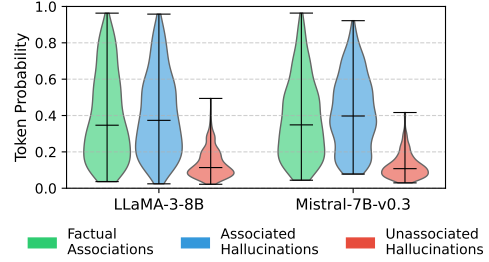


Figure 8: Distribution of last token probabilities.

the father of <subject> is”), resulting in significantly higher output entropy.

5 Revisiting Hallucination Detection

The mechanistic analysis in §4 reveals that *internal states of LLMs primarily capture how the model utilizes its parametric knowledge, rather than whether the output is truthful*. This suggests that representation-based signals struggle to distinguish associated hallucinations (AHs) from factual associations (FAs) but can more effectively separate unassociated hallucinations (UHs) from FAs.

In this section, we quantify the extent to which existing hallucination detection methods can separate AHs from FAs and compare this performance to detecting UHs.

Experimental Setups We revisit the effectiveness of widely-adopted white-box hallucination detection approaches that use internal state probing as well as black-box approaches that rely on scalar features. We evaluate on three settings: 1) **AH Only** (1,000 FAs and 1,000 AHs for training; 200 of each for testing), 2) **UH Only** (1,000 FAs and 1,000 UHs for training; 200 of each for testing), and 3) **Full** (1,000 FAs and 1,000 hallucination samples mixed of AHs and UHs for training; 200 of each for testing). For each setting, we repeat data splitting with five random seeds and report mean AUROC and standard deviation.

White-box methods: We extract and normalize

Methods	LLaMA		Mistral	
	AH Only	UH Only	AH Only	UH Only
Subject	0.65 $\pm$ 0.02	0.91 $\pm$ 0.01	0.57 $\pm$ 0.02	0.81 $\pm$ 0.02
Attention	0.58 $\pm$ 0.04	0.92 $\pm$ 0.02	0.58 $\pm$ 0.07	0.87 $\pm$ 0.01
Last Token	0.69 $\pm$ 0.03	0.93 $\pm$ 0.01	0.63 $\pm$ 0.02	0.92 $\pm$ 0.01
Probability	0.49 $\pm$ 0.01	0.86 $\pm$ 0.01	0.46 $\pm$ 0.00	0.89 $\pm$ 0.00
Subject Pop.	0.48 $\pm$ 0.01	0.87 $\pm$ 0.01	0.52 $\pm$ 0.01	0.84 $\pm$ 0.01

Table 2: Hallucination detection performance on **AH Only** and **UH Only** settings. Detailed statistical significance analysis is provided in Appendix E.1.

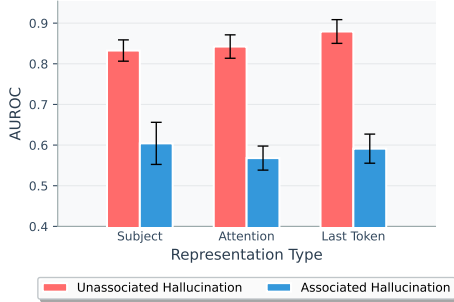


Figure 9: Hallucination detection performance on the **Full** setting (LLaMA-3-8B).

internal features and then train a probe.

- **Subject representations:** last subject token hidden state from three consecutive layers (Gottesman and Geva, 2024).
- **Attention flow:** attention weights from the last token to subject tokens across all layers (Yükselgönül et al., 2024).
- **Last-token representations:** final token hidden state from the last layer (Orgad et al., 2025).

Black-box methods: We test two commonly used scalar features, including **answer token probability** (Orgad et al., 2025) and **subject popularity** (average monthly Wikipedia page views) (Mallen et al., 2023a). As discussed in §4.2.2 and §4.4, these features are also related to whether the model relies on encoded knowledge to produce outputs rather than with truthfulness itself.

Experimental Results Table 2 shows that hallucination detection methods perform differently in **AH Only** and **UH Only** settings. For white-box probes, all approaches effectively distinguish UHs from FAs (AUROC $\approx$ 0.91–0.93). In contrast, performance drops sharply on the **AH Only** setting (AUROC $\approx$ 0.58–0.69). Black-box methods follow the same pattern (AUROC $\approx$ 0.86–0.87 on **UH Only** and $\approx$ 0.48–0.49 on **AH Only**). Notably, the performance of black-box methods on **AH Only** is close to random guessing (AUROC $\approx$ 0.48–0.49), indicating that token uncertainty and subject popularity provide little signal for separating AHs from

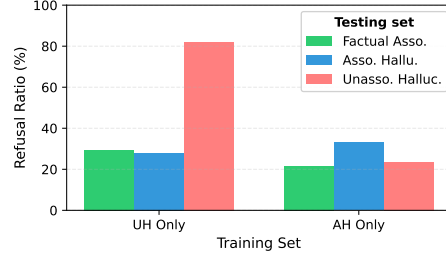


Figure 10: Refusal tuning performance across three types of samples (LLaMA-3-8B).

FAs. Although white-box representations show slightly better separability than black-box methods, such performance is far from reliable for deployment (Hosmer Jr et al., 2013). Figure 9 further highlights this disparity under the **Full** setting: detection is consistently stronger on UH samples than on AH samples, and adding AHs to the training set significantly dilutes performance on UHs (AUROC $\approx$ 0.9 on **UH Only** vs. $\approx$ 0.8 on **Full**).

These results confirm that both internal probes and black-box methods capture whether a model draws on parametric knowledge, not whether its outputs are factually correct. Our further analysis in Appendix E.2 reveals that consistency-based detection methods also struggle to distinguish AHs from FAs. UHs are easier to detect because they bypass this knowledge, while associated hallucinations are produced through the same recall process as FAs, leaving limited internal cues to distinguish them. An analysis of sensitivity-specificity trade-offs in Appendix E.3 further confirms this limitation, showing that while UHs can be detected with high sensitivity and specificity simultaneously, AH detection requires a significant trade-off between them. As a result, **LLMs appear to have limited intrinsic awareness of their own truthfulness, and detection methods relying on these signals risk misclassifying AHs as correct**, fostering harmful overconfidence in model outputs.

6 Challenges of Refusal Tuning

A common strategy to mitigate potential hallucinations in the model’s responses is to fine-tune LLMs to refuse answering when they cannot provide a factual response, e.g., **Refusal Tuning** (Zhang et al., 2024). For such refusal capability to generalize, the training data must contain a shared feature pattern across hallucinated outputs, allowing the model to learn and apply it to unseen cases.

Our analysis in the previous sections shows that

this prerequisite is not met. Because AH representations are geometrically diverse, it is difficult for models to capture a consistent feature pattern among AH samples, thereby hindering the generalization of refusal responses to unseen AH samples. In contrast, UH representations tend to form distinct cluster, presenting consistent feature patterns that should facilitate easier generalization of the refusal capability.

Experimental Setups To verify the hypothesis, we conduct refusal tuning on LLMs under two settings: 1) **UH Only**, where 1,000 UH samples are paired with 10 refusal templates, and 1,000 FA samples are preserved with their original answers. 2) **AH Only**, where 1,000 AH samples are paired with refusal templates, with 1,000 FA samples again remain unchanged. We then evaluate both models on 200 samples each of FAs, UHs, and AHs. A response matching any refusal template is counted as a refusal, and we report the **Refusal Ratio** as the proportion of samples eliciting refusals. This measures not only whether the model refuses appropriately on UHs and AHs, but also whether it “over-refuses” on FA samples.

Experimental Results Figure 10 shows that training with UHs leads to strong generalization across UHs, with refusal ratios of 82% for LLaMA. However, this effect does not transfer to AHs, where refusal ratios fall to 28%, respectively. Moreover, some FA cases are mistakenly refused (29.5%). These results confirm that UHs share a common activation subspace, supporting generalization within the category, while AHs and FAs lie outside this space. By contrast, training with AHs produces poor generalization. On AH test samples, refusal ratio is only 33%, validating that their subject-specific hidden states prevent consistent refusal learning. Generalization to UHs is also weak (23.5%), again reflecting the divergence between AH and UH activation spaces.

Overall, these findings indicate that **the generalizability of refusal tuning is constrained by the heterogeneous nature of hallucinations**. While UH representations are internally consistent enough to support learnable refusal generalization, AH representations are too geometrically diverse for the model to learn a consistent and robust refusal pattern.

7 Conclusions and Future Work

In this work, we revisit the widely accepted claim that hallucinations can be detected from a model’s internal states. Our mechanistic analysis reveals that **hidden states encode whether models rely on their parametric knowledge rather than truthfulness**. As a result, detection methods succeed most reliably when outputs are detached from the input but struggle when hallucinations arise from the same knowledge-recall process as correct answers.

These findings lead to three key implications. First, future evaluations should **report detection performance separately** for Associated Hallucinations (AHs) and Unassociated Hallucinations (UHs), as they stem from fundamentally different internal processes and require distinct detection strategies. Second, relying solely on hidden states is insufficient for reliable hallucination detection. Future research should integrate LLMs with **external feedback mechanisms**, such as fact-checking modules or retrieval-based verifiers, to assess factuality more robustly. Third, future studies should **prioritize improving AH detection**. Because AHs occur more frequently in widely known or highly popular topics (§4.2), their undetected errors pose greater risks to user trust and the practical reliability of LLMs.

Limitations

We identify several limitations of our work.

Focus on Factual Knowledge While our analysis identifies challenging cases of hallucination detection methods, our study is primarily limited to factual completion prompts. It does not extend to long-form or open-ended text generation tasks (Wei et al., 2024; Min et al., 2023; Huang and Chen, 2024). Future work should broaden this investigation to these tasks in order to draw more comprehensive conclusions.

Lack of Analysis on Prompt-based Hallucination Detection Approaches Our analysis focuses on white-box hallucination detection methods based on internal states and two black-box approaches based on external features. We do not include verbalization-based strategies (Lin et al., 2022a; Tian et al., 2023; Xiong et al., 2024; Yang et al., 2024b; Ni et al., 2024; Zhao et al., 2024), such as prompting the model to report or justify